



as easily as they can run up them; and their young are able to keep up with them when just a few weeks old.

Coping with winter

In tropical mountains, conditions are often much the same all year-round, which means that animals can stay at one elevation all their lives. But in temperate mountains, seasonal changes affect the food supply. Winter is the critical time: anything that cannot survive the cold weather conditions and the shortage of food has to move to lower ground or hibernate until the return of spring.

Animals that are resident at high elevation have a variety of ways of dealing with the changes. Insects often enter a dormant state, called diapause, which puts their development on hold. Many small mammals, such as marmots, survive mountain winters by hibernating, while many of those that remain active live on food stores accumulated earlier in the year. Pikas, for example, gather up leaves and grass and build them into “haystacks” among the broken rocks around their homes. Before adding fresh supplies to a stack, they sometimes spread them out to dry in the sunshine, which reduces the chances of the food from rotting.

For other animals, the first fall snows are the signal to move downhill. These vertical migrations are a common feature of mountain

VERTICAL MIGRANTS

In mountainous regions, red deer spend the summer high up, where food is plentiful and there are relatively few biting flies. Their downhill migration in fall often coincides with the start of the rutting season, when males grow a mane of hair on the neck and compete with each other for the right to mate.

life in temperate regions, and they are demonstrated by a wide range of mammals and birds, from mountain sheep and deer to choughs and grouse. In many cases, the migration involves moving from the exposed

mountaintop to the forests lower down, but some mountain forest species also migrate. Among these latter migrants are birds such as nutcrackers, which feed on conifer seeds. If the seed crop fails, they fly downhill in a form of sporadic migration called an irruption (see p.59). Clark’s nutcracker, from the Rocky Mountains, is a typical example: normally found at up to 8,200 ft (2,500 m), it descends as low as sea level when food becomes hard to find.

Finding food

As in most land habitats, a mountain’s animal life depends ultimately on plants, for plants provide food for herbivorous animals, which, in turn, are eaten by a wide range of predators. However, some mountain animals make use of a very different food source—the cargo of small animals, mainly insects, that are carried uphill by the wind to be stranded among the rocks, snow, and ice.

Most of these wind-blown animals are so tiny that they are practically invisible; yet they provide useful nourishment for scavengers that live above the snow line. They consist almost entirely of invertebrates, such as springtails and snow fleas, which can survive the very low temperatures of high-altitude winters. During the depths of winter, they hide among rocks and moss, but when the weather warms up, they can often be seen hopping across banks of snow, feasting on the debris that the wind has brought up from lower ground.

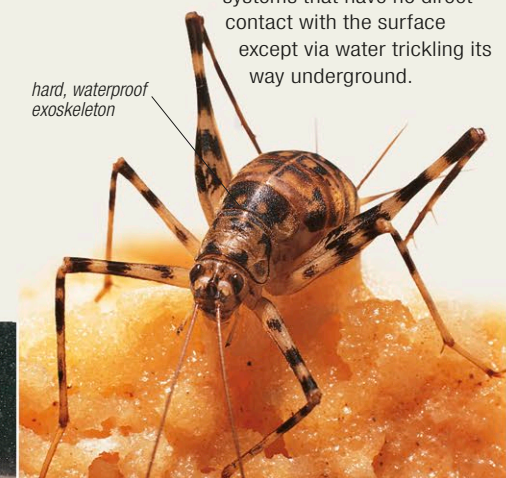
LIFE IN CAVES

Many animals use caves temporarily, but some have adapted to spend their entire lives in them. These permanent cave dwellers, called troglodytes, feed either on each other or on the droppings deposited by roosting bats and birds.

As a habitat, deep caves have the advantage that temperatures remain fairly constant throughout the year. However, they are also completely dark, which means that eyes are useless. Bats, oilbirds, and swiftlets use echolocation to navigate while underground, but permanent cave dwellers sense their surroundings largely by touch, often using smell to track down food. Cave crickets detect food using antennae, while spiders and harvestmen use their feet. In subterranean streams and pools, cave salamanders, such as the olm, sense vibrations in

the water. The cave fish has a row of pressure sensors along each side of its body that enables it to detect other animals several yards away.

Although food is scarce, animals exist many miles underground and have even been found in pothole systems that have no direct contact with the surface except via water trickling its way underground.



GIANT ANTENNAE

Cave crickets’ antennae, which can be 2 or 3 times the body length, locate the dead remains and bat droppings that make up the animal’s diet.

UNDERWATER LIFE

Like most aquatic vertebrates in caves, the olm has very little pigment in its skin. Its eyes are vestigial and covered with skin. Olms breed in water and remain in it throughout adult life.



FLEXIBLE MIGRANT

Like many temperate mountain animals, the nutcracker lives at various altitudes, depending on the conditions: severe cold drives it downhill, but because there is more competition for food on low ground it returns uphill as soon as the weather improves.

